# Week – 4 Assignment

## Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

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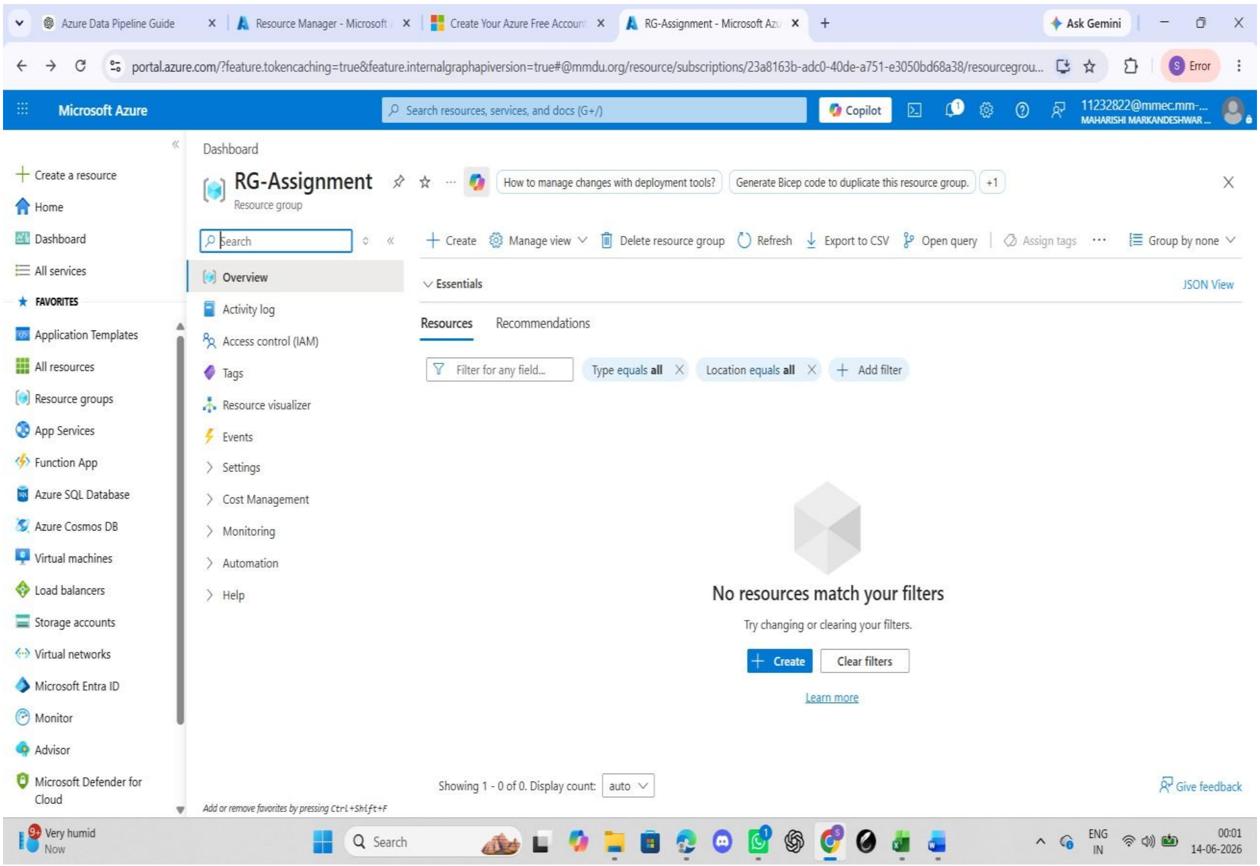
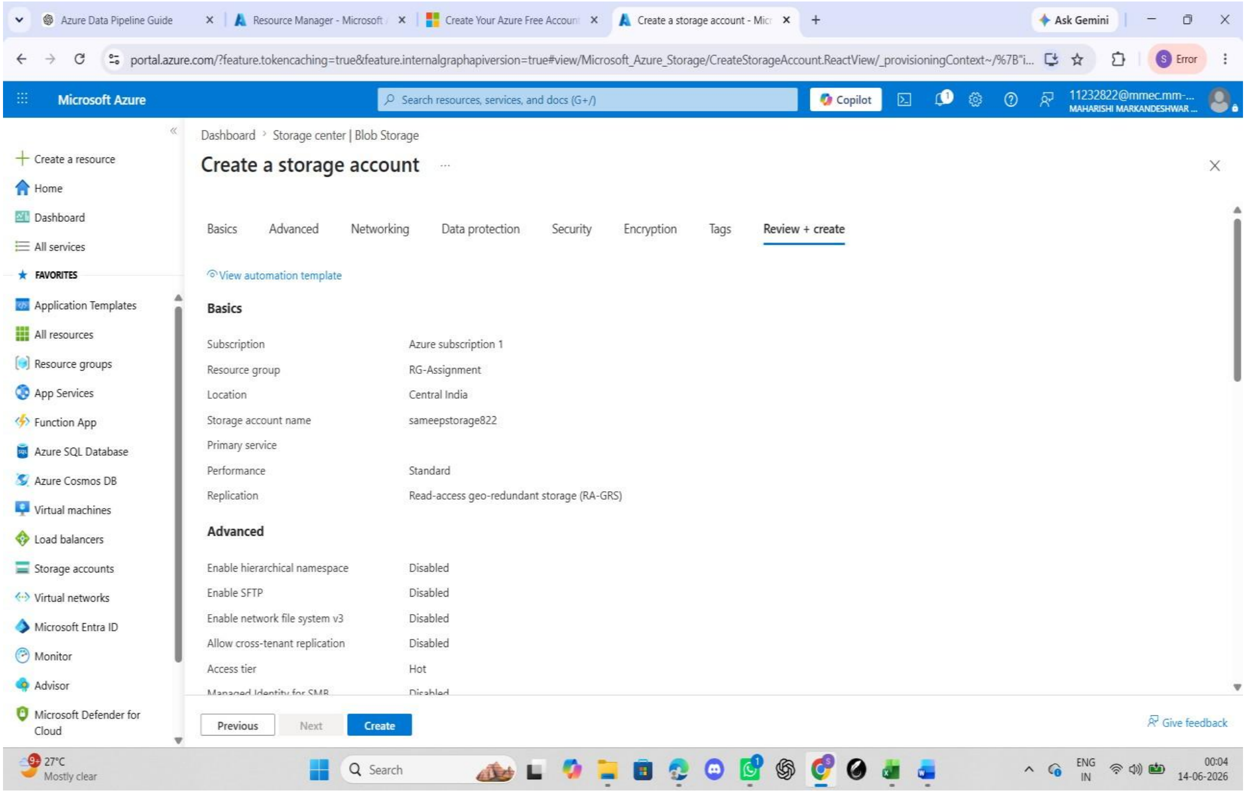
## 1. Objective

Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory. Steps: Explore Azure Portal and create a Resource Group. Create a Storage Account, Blob Container, and upload a CSV file. Create Azure Data Factory and explore ADF UI (Author, Monitor, Manage). Create Linked Service (Blob Storage) and datasets (source and destination). Use Get Metadata activity to retrieve file information. Build a pipeline using Copy Data activity and configure source and destination. Execute the pipeline using Debug/Trigger and monitor execution. Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage. Implement an end-to-end pipeline (Blob → ADF → Destination) with metadata validation.

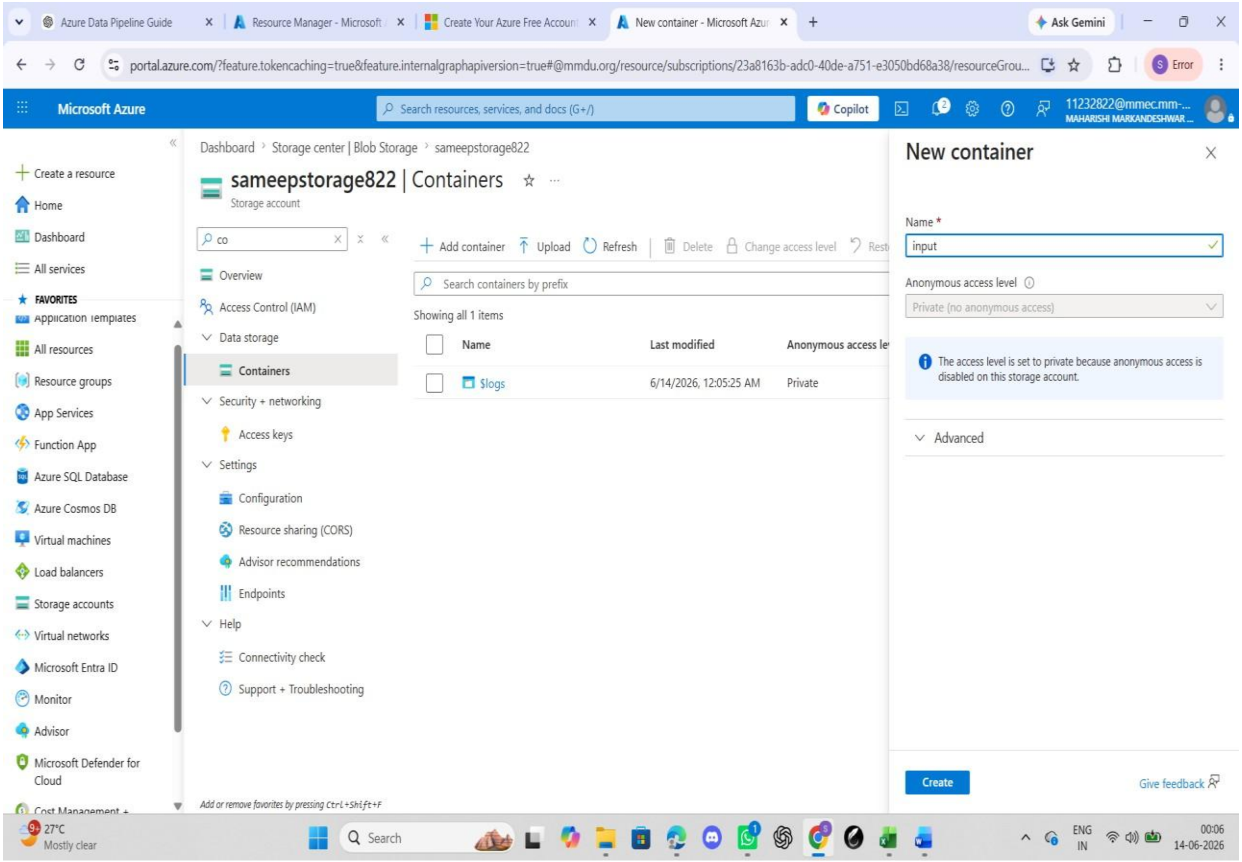
## 2. Resource Group Creation



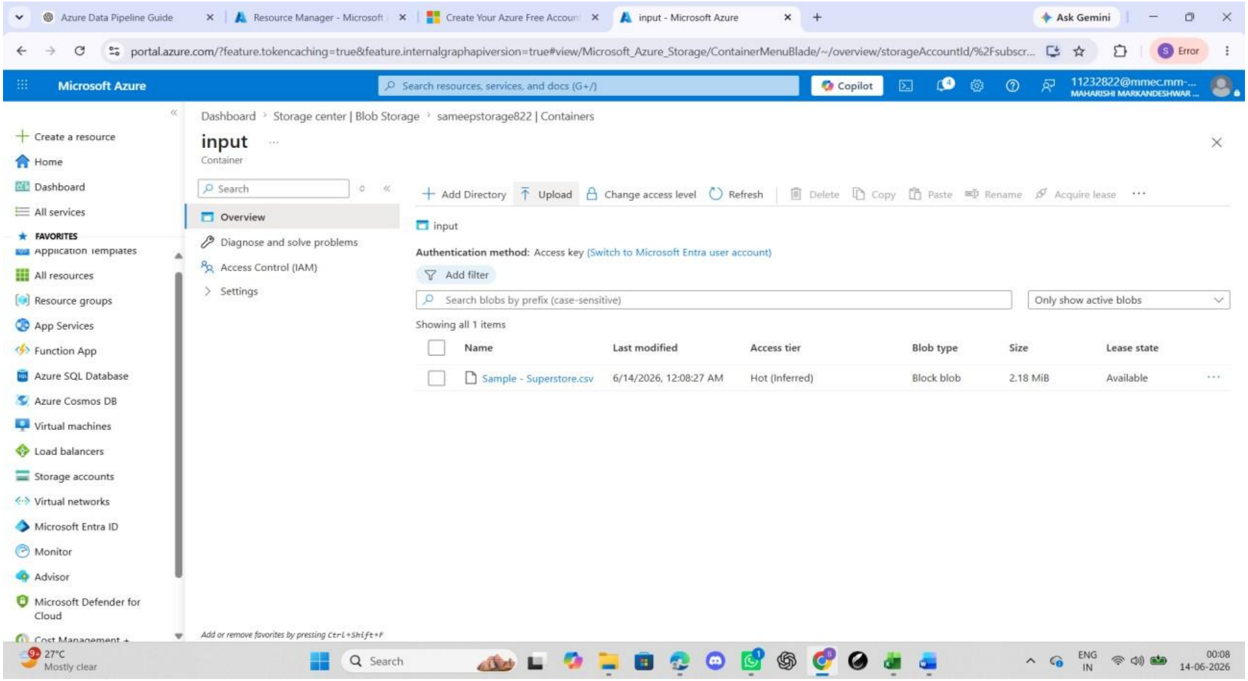
# 3. Storage Account Creation



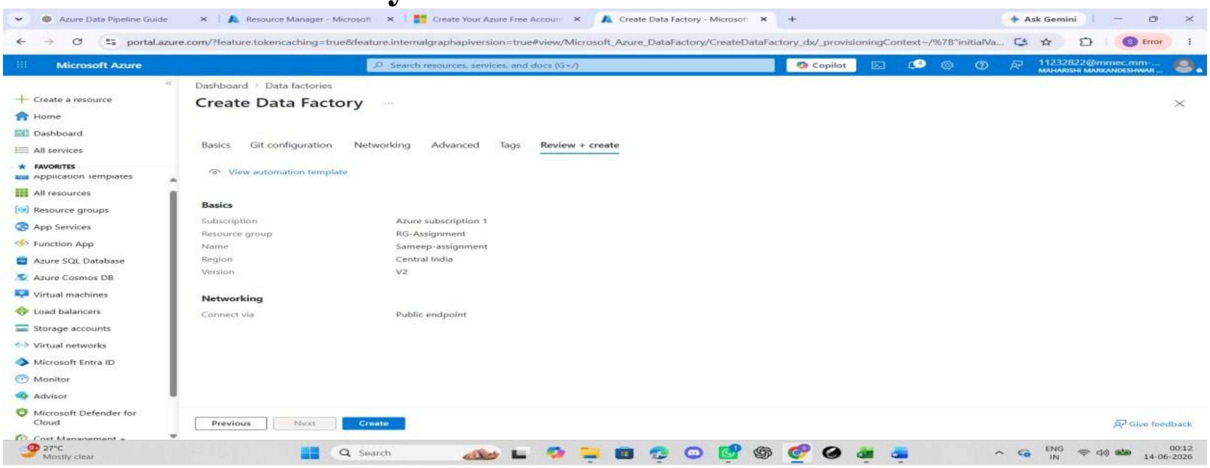

# 4. Blob Container Creation

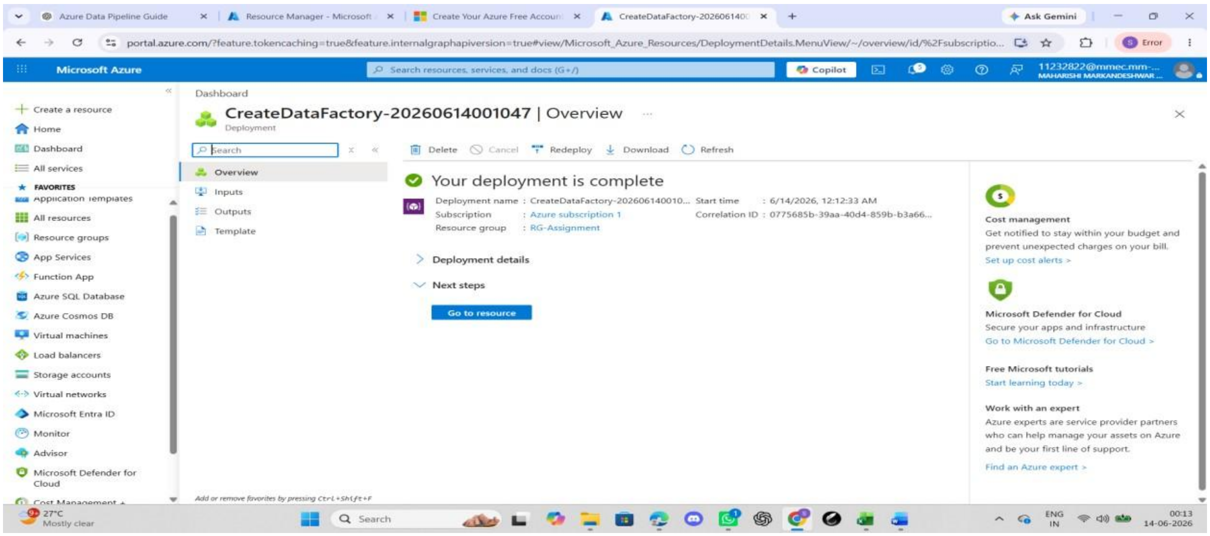


# 5. CSV File Upload

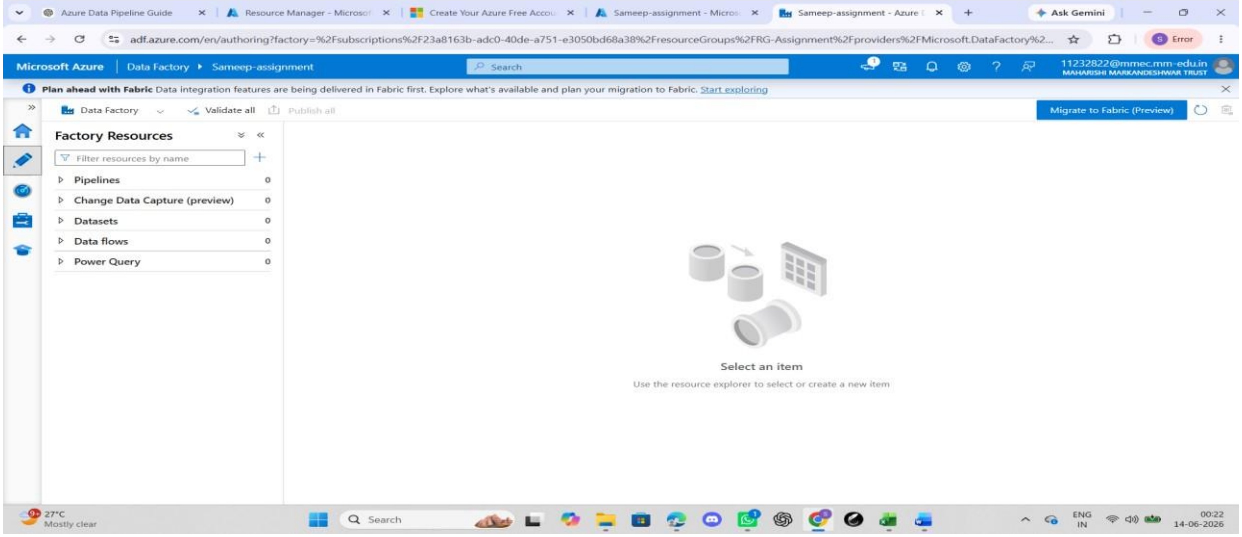


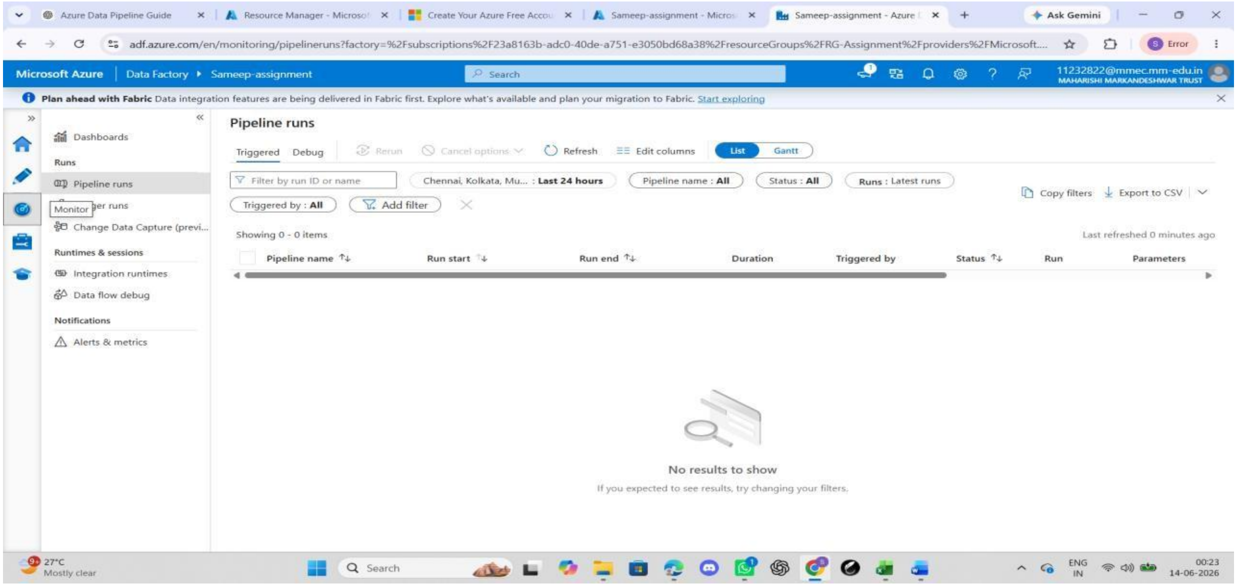

# 6. Azure Data Factory Creation

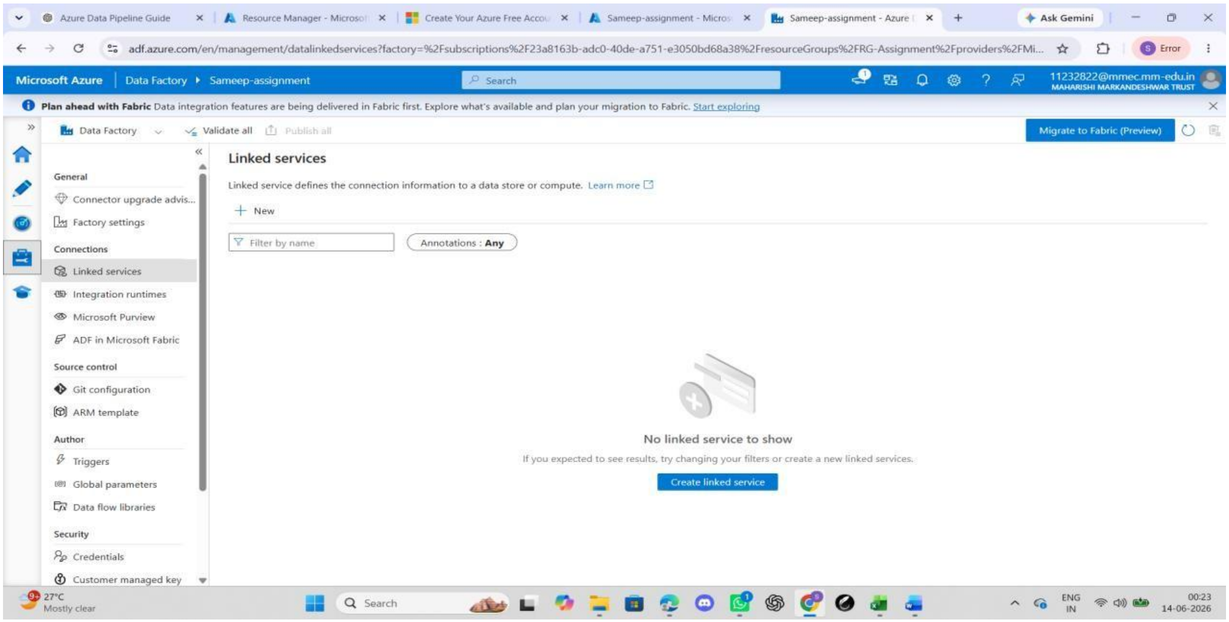


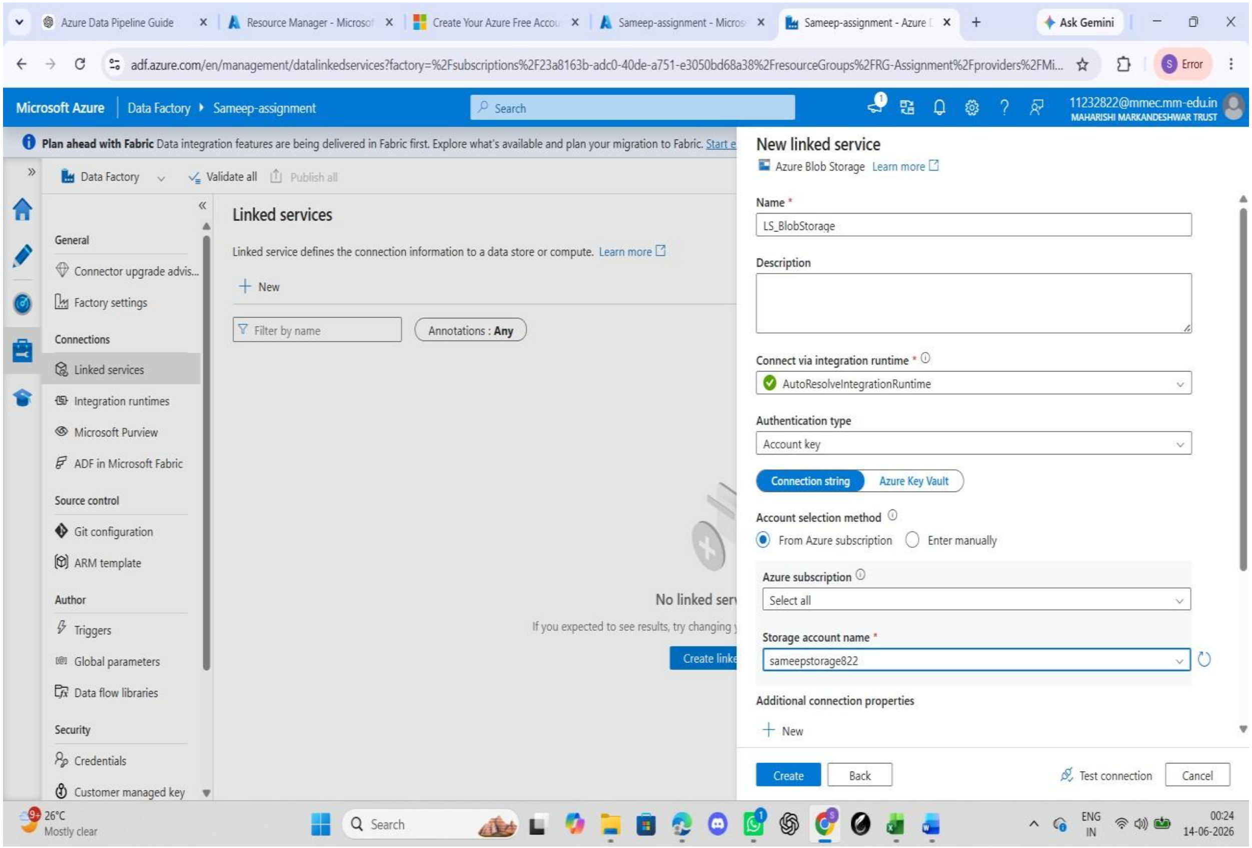
# 7. ADF Studio Overview



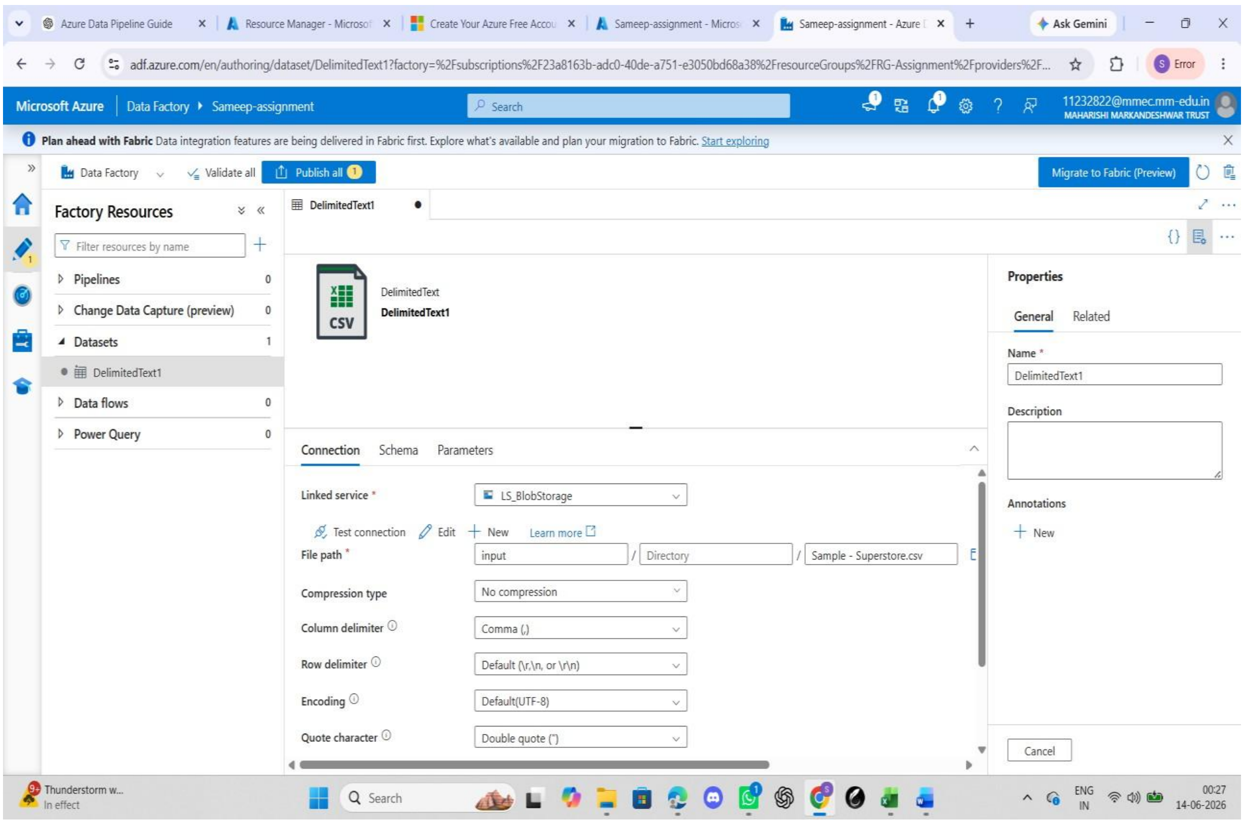



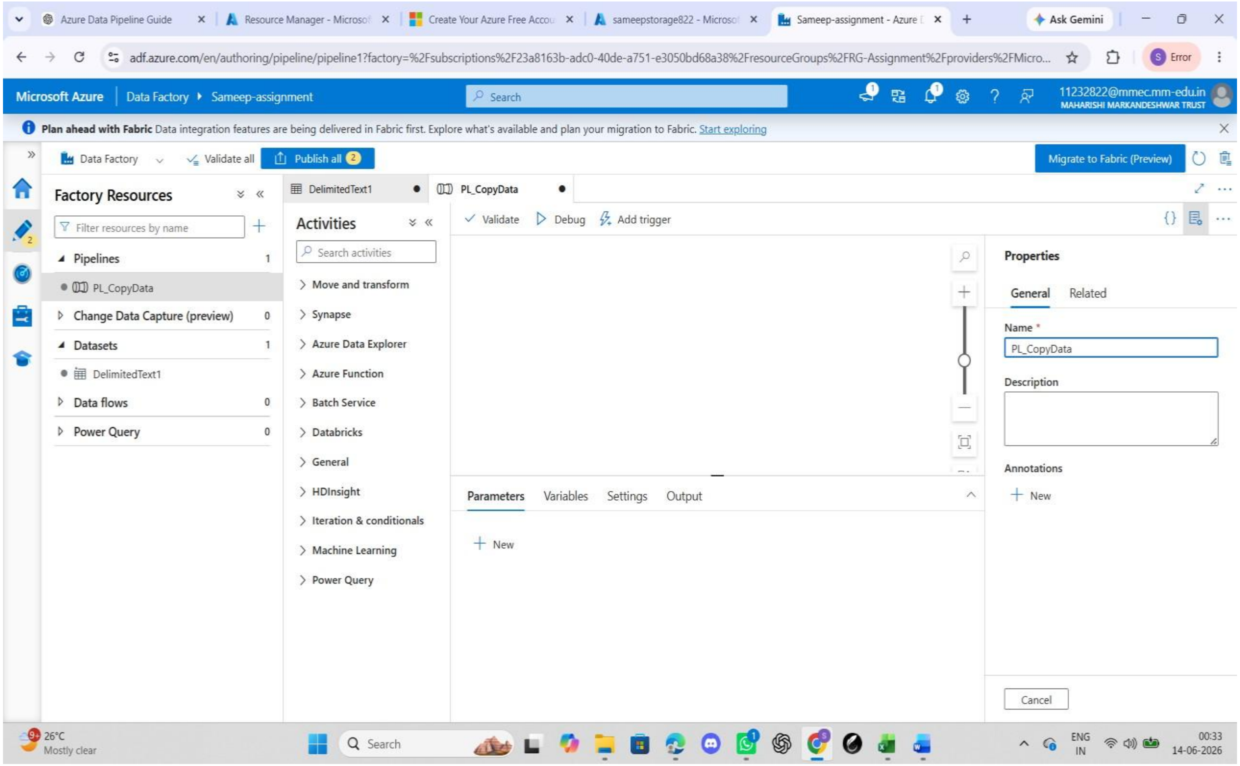
# 8. Linked Service Configuration



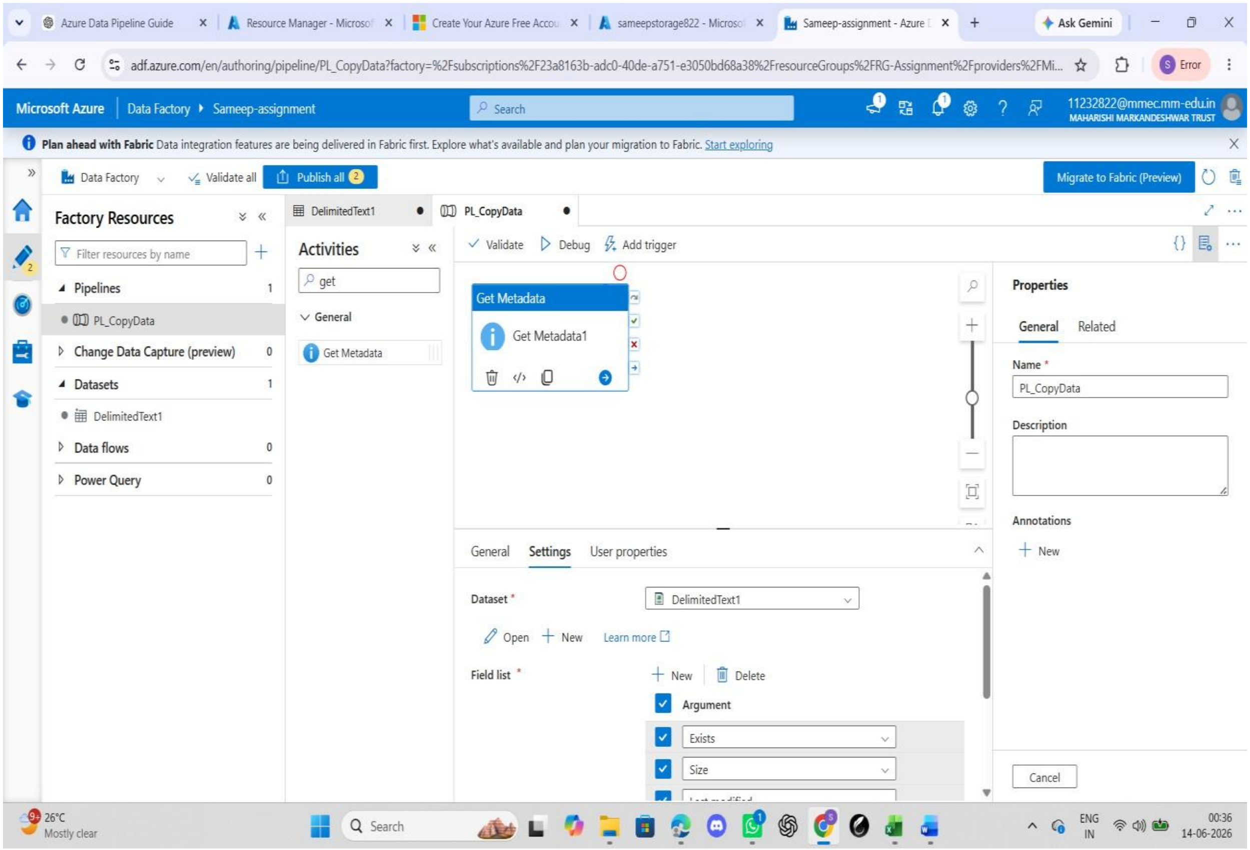

# 9. Source Dataset Creation



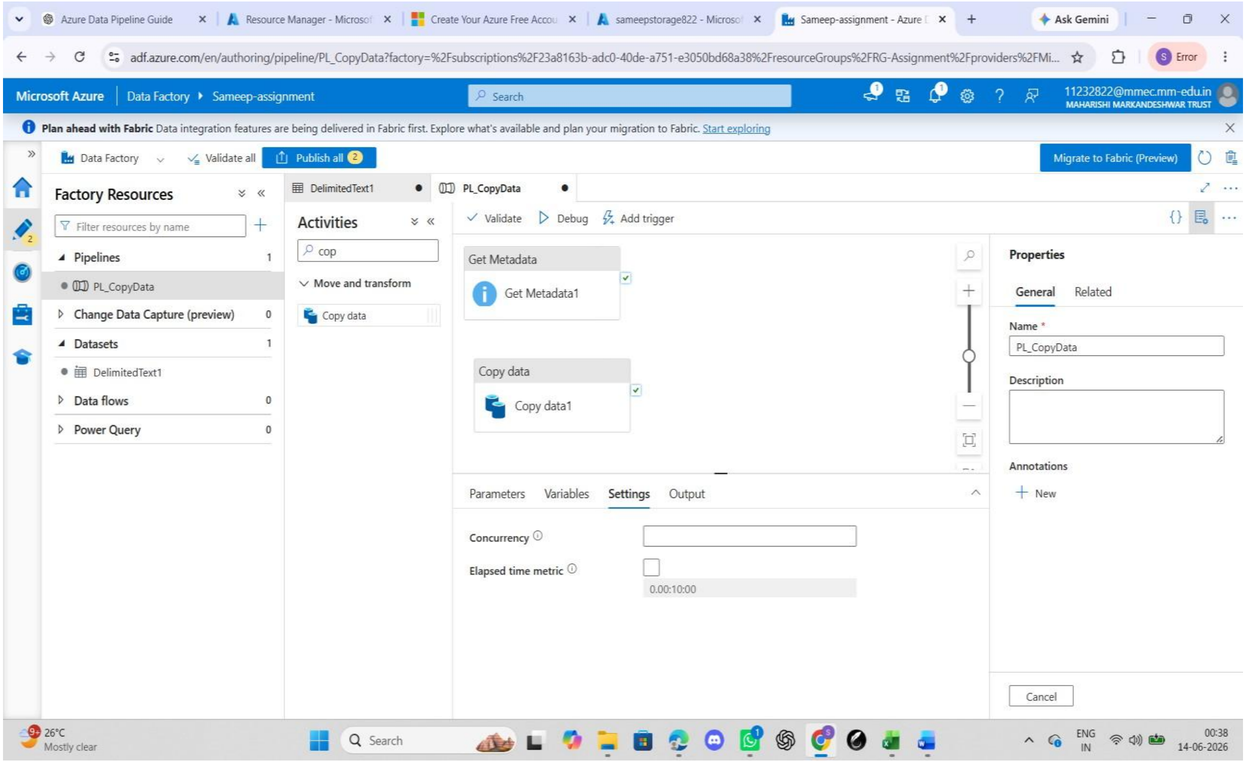
# 10. Pipeline Creation



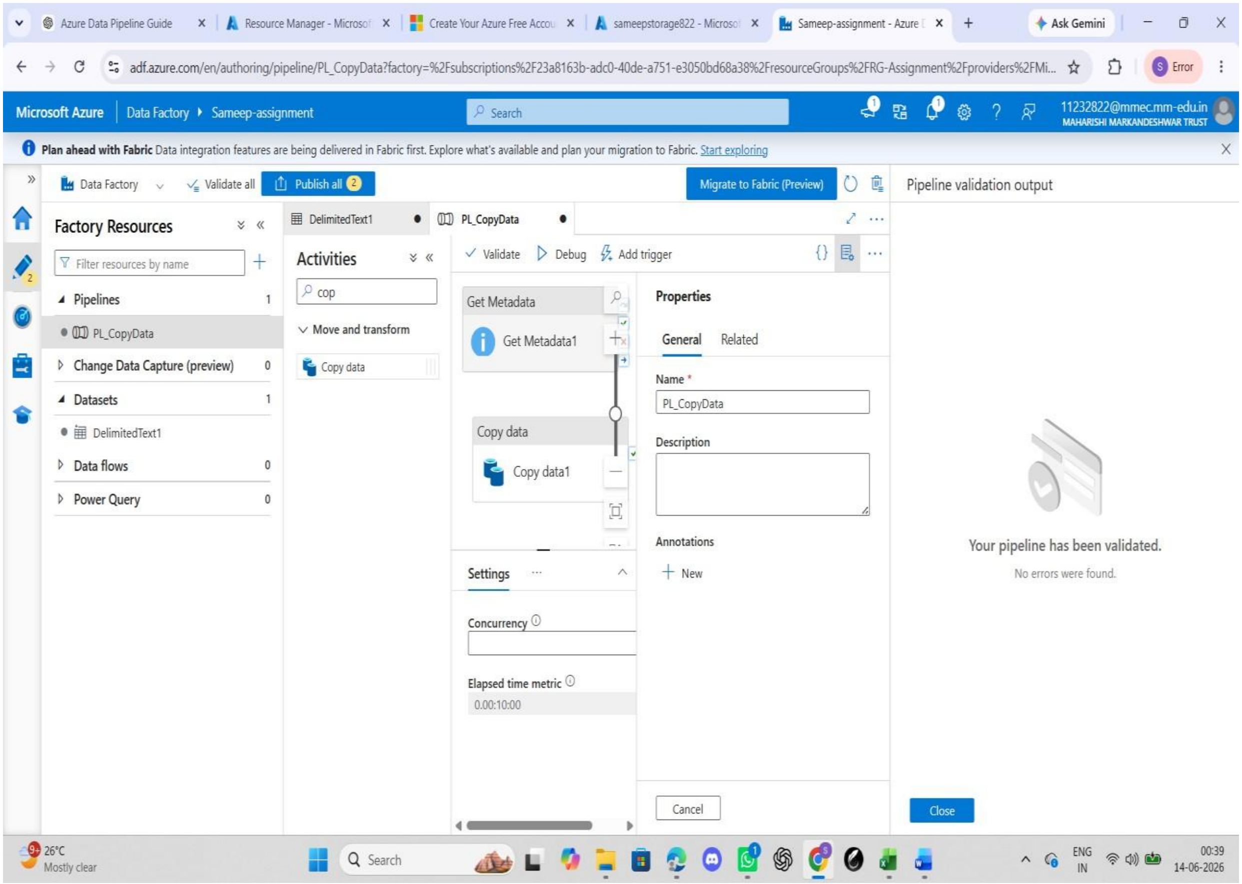
# 11. Get Metadata Activity



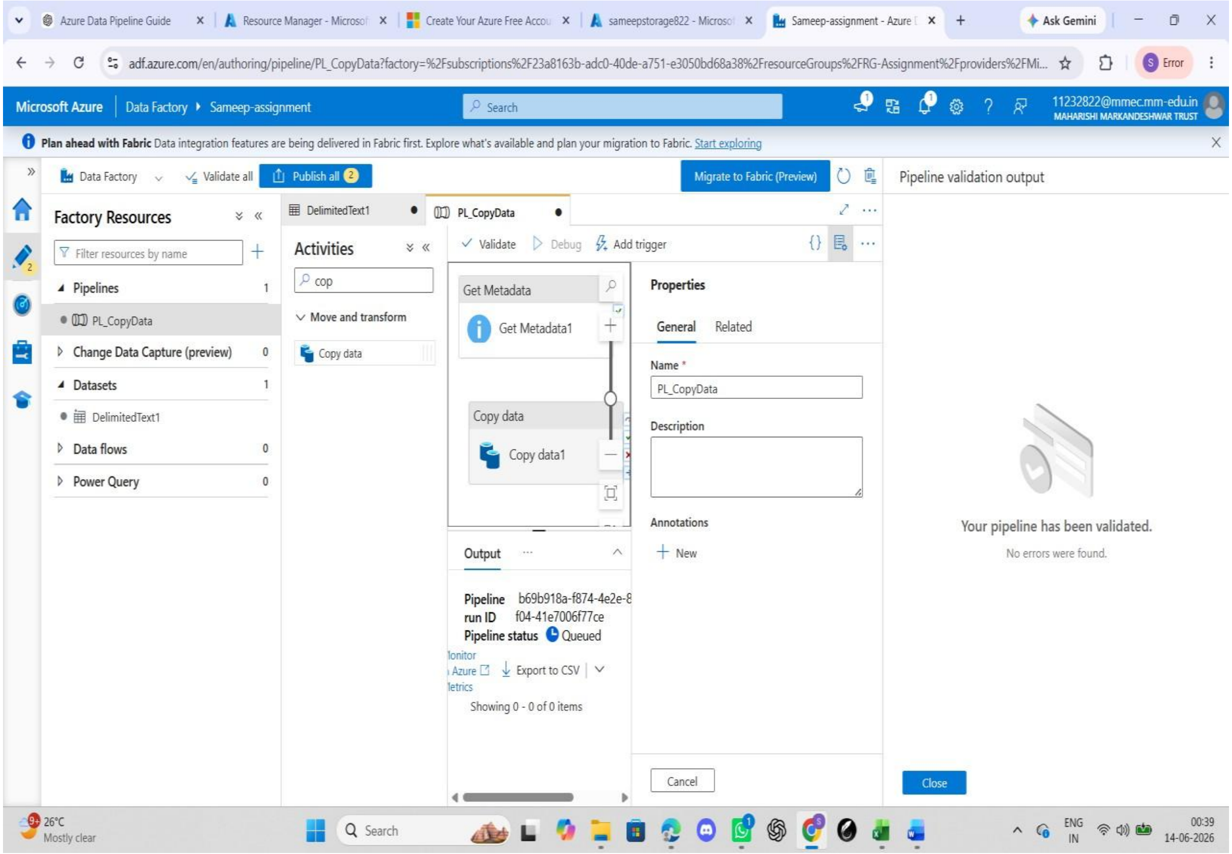
# 12. Copy Data Activity



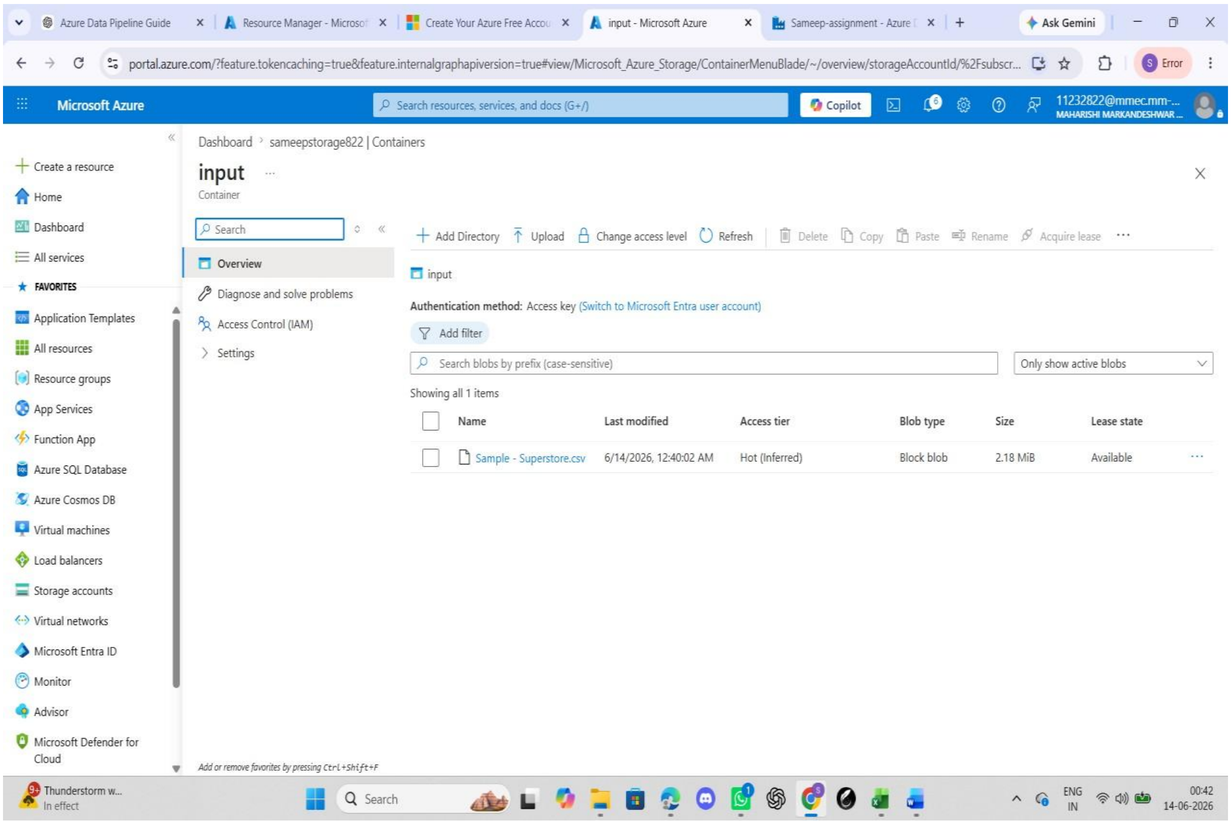

# 13. Validate



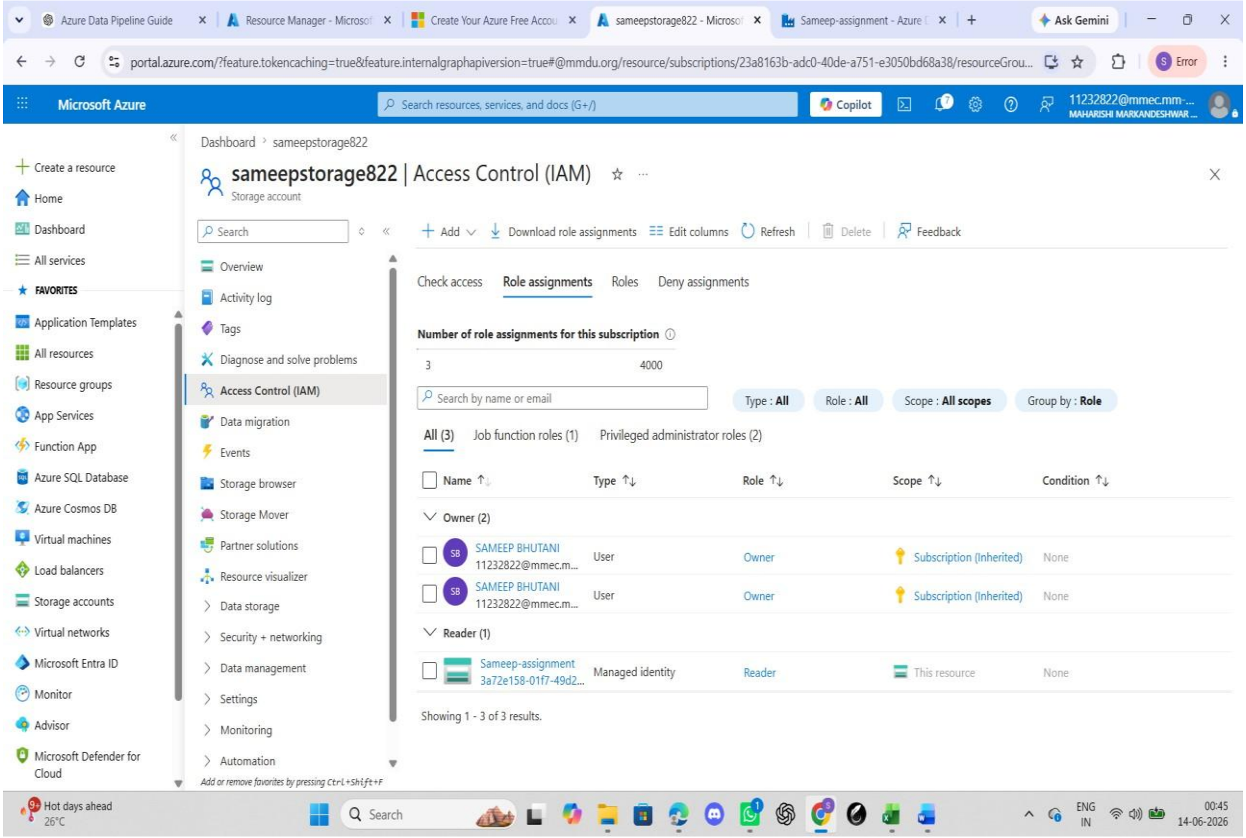
# 14. Debug




# 15. Output Check



## 16. IAM Role Assignment




## 17. Result

The pipeline executed successfully and copied the CSV file from the source container to the destination container.

## 18. Conclusion

This assignment successfully demonstrated the implementation of an end-to-end data pipeline using Microsoft Azure services. A Resource Group, Storage Account, Blob Containers, and Azure Data Factory were created and configured. The provided CSV file was uploaded to Azure Blob Storage and connected to Azure Data Factory through Linked Services and Datasets. The Get Metadata activity was used to validate and retrieve information about the source file, while the Copy Data activity was used to transfer the file from the source container to the destination container. The pipeline was executed successfully using the Debug and Trigger options, and the execution status was monitored through the Monitor section of Azure Data Factory. Proper access permissions were also ensured through IAM role assignments. The successful creation and execution of the pipeline confirmed the seamless integration between Azure Storage and Azure Data Factory for data movement and processing.